

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Reusable plug, Part Number 4918874
- Gasket scraper or cleaning pad, Part Number 3823258.

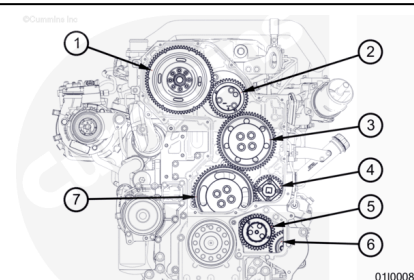
Additional Service Items

- No additional service items required.

General Information

This illustration depicts the gear housing with the gear covers removed for gear trains with a solid style lower idler gear.

1. Camshaft gear, valve
2. Idler gear, adjustable
3. Idler gear, upper
4. Accessory gear
5. Idler gear, fuel pump
6. Fuel pump gear
7. Idler gear, lower solid.



The crankshaft gear is **not** shown and does **not** need to be removed in order to remove the gear housing.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

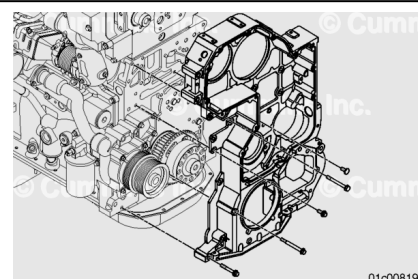
Note : Before removing gears, read the Engine Base Timing procedure. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-088.html>)

- Disconnect the batteries. See the equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html>)
- Remove the water pump drive belt. Refer to Procedure 008-003 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-003.html>)
- Remove the alternator drive belt. Refer to Procedure 013-005 in Section 13. (</qs3/pubsys2/xml/en/procedures/377/377-013-005.html>)
- Remove the fan hub assembly. Refer to Procedure 008-036 in Section 8. (</qs3/pubsys2/xml/en/procedures/377/377-008-036-tr.html>)
- Remove the alternator/refrigerant compressor mounting bracket. Refer to Procedure 009-056 in Section 9. (</qs3/pubsys2/xml/en/procedures/377/377-009-056.html>)
- Remove the crankcase breather and tube. Refer to Procedure 003-001 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-001.html>)
- Remove the vibration damper(s). Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-052.html>)
- Remove the crankshaft position sensor. Refer to Procedure 019-365 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-365.html>)
- Remove the upper gear cover. Refer to Procedure 001-079 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-079.html>)
- Remove the lower gear cover. Refer to Procedure 001-080 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-080.html>)
- Remove the camshaft position sensor. Refer to Procedure 019-363 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-363.html>)
- Remove the valve camshaft gear, adjustable idler gear, and upper idler gear. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-088.html>)
- Remove the camshaft retainer plate. Refer to Procedure 002-050 in Section 2. (</qs3/pubsys2/xml/en/procedures/377/377-002-050.html>)
- Remove the adjustable idler mounting spacer. Refer to Procedure 001-077 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-077.html>)

- Remove the air compressor, if applicable. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/377/377-012-014.html>)
- Remove the fuel pump idler gear and the lower idler scissor gear. Refer to Procedure 001-088 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-088.html>)
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-016.html>)
- Remove the capscrews from the front of the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)
- Remove the retaining brace and bracket. Refer to Procedure 001-081 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-081.html>)
- Remove the accessory bracket. Refer to Procedure 001-082 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-082.html>)
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-023.html>)
- If equipped, remove the lubricating oil fill tube. Refer to Procedure 007-065 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-065.html>)
- Remove the fuel supply line from the fuel pump to the fuel rail. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- Remove the front engine support bracket. Refer to Procedure 016-002 in Section 16. (</qs3/pubsys2/xml/en/procedures/377/377-016-002.html>)

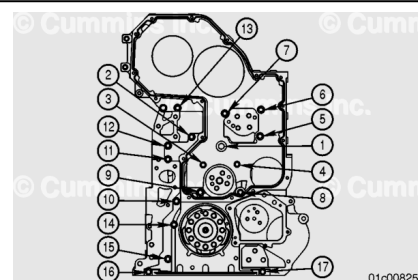
Remove

Remove two capscrews and install two guide studs.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

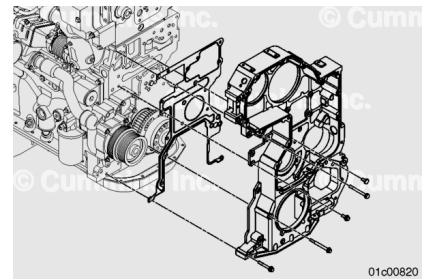


Remove and discard capscrew(s) 2 through 13.

- Captive washer capscrew M10 x 1.5 x 25.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the remaining two capscrews.

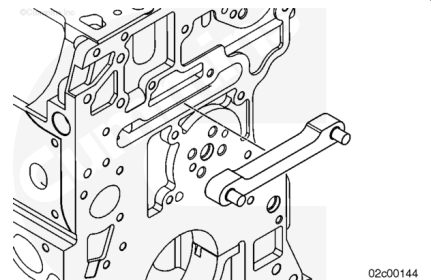
Remove the housing and gasket.

Discard the gasket.

Remove the guide studs.

⚠ CAUTION ⚠

Immediately upon removal of the front gear housing, a plug must be inserted into the cylinder block oil passage drilling. Failure to insert the oil passage plug can result in a bearing failure, crankshaft failure, or both.



The reusable plug, Part Number 4918874, is necessary to prevent debris from entering the lubrication system during the repair.

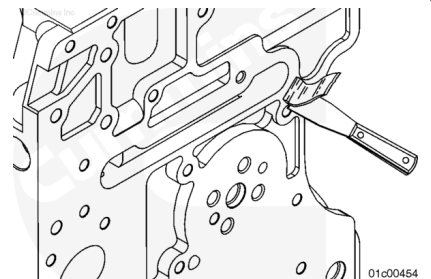
Insert the plug into the front oil passage.

Firmly push the plug into the oil passage to prevent debris from entering the lubricating oil system when cleaning the gasket surface.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

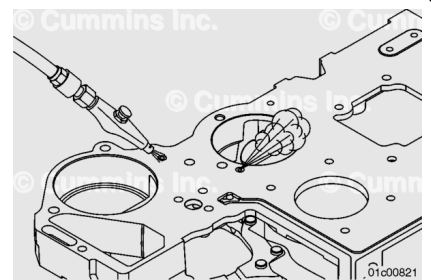
If the oil passages are not plugged before cleaning, a crankshaft failure can occur.

Clean the gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258. If the gasket material residue can **not** be felt with a finger, the surface is ready to accept the new gasket.

Use solvent to clean the gear housing.

Dry with compressed air.

Use compressed air to clear the oil galley to the air compressor and fuel pump of any obstructions.



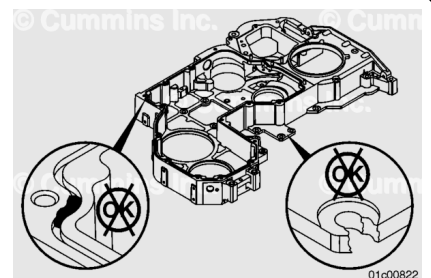
Inspect the gear housing for cracks or other damage.

Inspect the seal grooves of the housing for debris or damage.

If the gear housing is cracked or otherwise damaged, it **must** be replaced.

⚠ CAUTION ⚠

If an incompatible gear housing and fuel pump idler shaft are assembled together engine damage can occur.



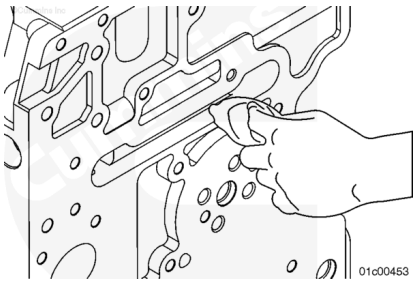
If replacing the front gear housing with a new part, ensure that the fuel pump idler is compatible with the new gear housing.

Install

Remove the protective plug from the cylinder block oil passage.

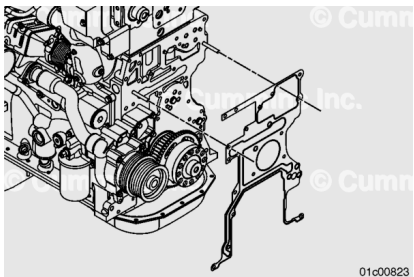
Make sure **no** debris enters the lubricating oil system.

Use a clean shop towel to carefully wipe out the oil passage.
Inspect the oil passages to make sure **no** debris is left.

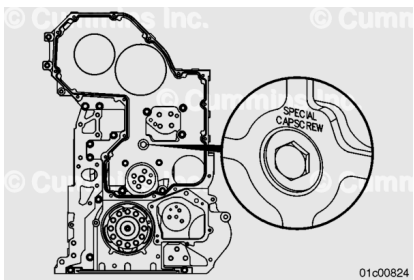


Install two guide studs.

Install a new gear housing mounting gasket.

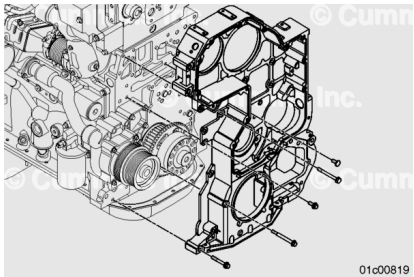


The gear housing has the words “Special Capscrew” cast into the housing at the mounting location (see illustration). This capscrew **must** be the short head capscrew to prevent contact between the capscrew and the idler gear.



⚠ WARNING ⚠

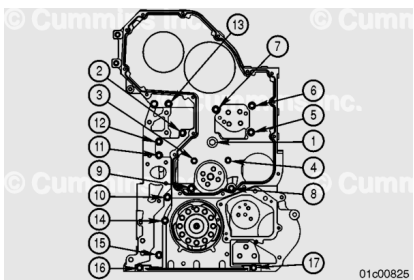
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the gear housing.

Install the mounting capscrews.

Four different length capscrews are utilized on the front gear housing. Use the table below to identify the capscrews and their mounting locations.



Capscrew Mounting Location		
Location	Size	Quantity
14	M10 x 1.5 x 100	1
15, 16, 17	M10 x 1.5 x 70	3

Capscrew Mounting Location		
Location	Size	Quantity
1	M10 x 1.5 x 21.8 (Special Short Head)	1
All remaining	M10 x 1.5 x 25	12

Remove the guide studs and install the two remaining capscrews.

Tighten the capscrews in the sequence shown.

Torque Value:

All Capscrews Except (1)

1. 20 N•m [177 in-lb]
2. 64 N•m [47 ft-lb]
3. Rotate capscrews an additional 30 degrees

Torque Value:

Capscrew (1)

1. 20 N•m [177 in-lb]
2. 56 N•m [41 ft-lb]

Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the front engine support bracket. Refer to Procedure 016-002 in Section 16. (</qs3/pubsys2/xml/en/procedures/377/377-016-002.html>)
- Install the fuel supply line from the fuel pump to the fuel rail. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- If equipped, install the lubricating oil fill tube. Refer to Procedure 007-065 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-065.html>)
- Install the front crankshaft seal. Refer to Procedure 001-023 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-023.html>)

- Install the accessory bracket. Refer to Procedure 001-082 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-082.html)
- Install the retaining brace bracket. Refer to Procedure 001-081 in Section 1
(/qs3/pubsys2/xml/en/procedures/377/377-001-081.html)
- Install the capscrews at the front of the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/377/377-007-025.html)
- Install the fuel pump. Refer to Procedure 005-016 in Section 5.
(/qs3/pubsys2/xml/en/procedures/377/377-005-016.html)
- Install the lower idler scissor gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
- Install the fuel pump idler gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
- Install the air compressor, if applicable. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/377/377-012-014.html)
- Install the camshaft retainer plate. Refer to Procedure 002-050 in Section 2.
(/qs3/pubsys2/xml/en/procedures/377/377-002-050.html)
- Install the adjustable idler mounting spacer. Refer to Procedure 001-077 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-077.html)
- Install the upper idler gear. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
- Install the adjustable idler and the valve camshaft gears. Refer to Procedure 001-088 in Section 1. (/qs3/pubsys2/xml/en/procedures/377/377-001-088.html)
- Install the camshaft position sensor. Refer to Procedure 019-363 in Section 19.
(/qs3/pubsys2/xml/en/procedures/377/377-019-363.html)
- Install the upper gear cover. Refer to Procedure 001-079 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-079.html)
- Install the lower gear cover. Refer to Procedure 001-080 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-080.html)
- Install the crankshaft position sensor. Refer to Procedure 019-365 in Section 19.
(/qs3/pubsys2/xml/en/procedures/377/377-019-365.html)
- Install the vibration damper(s). Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/377/377-001-052.html)
- Install the crankcase breather and tube. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/377/377-003-001.html)
- Install the alternator/refrigerant compressor mounting bracket. Refer to Procedure 009-056 in Section 9. (/qs3/pubsys2/xml/en/procedures/377/377-009-056.html)
- Install the fan hub assembly. Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/377/377-008-036-tr.html)
- Install the alternator drive belt. Refer to Procedure 013-005 in Section 13.
(/qs3/pubsys2/xml/en/procedures/377/377-013-005.html)
- Install the water pump drive belt. Refer to Procedure 008-003 in Section 8.
(/qs3/pubsys2/xml/en/procedures/377/377-008-003.html)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/377/377-008-018-tr.html)
- Connect the batteries. See the equipment manufacturer service information.

- Operate the engine to normal operating temperature. Check for leaks.

Last Modified: 16-Nov-2023
